

HAINES, AK, USA

WMO: 703626

Lat: 59.243N

Lon: 135.509W

Elev: 5

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25323

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-14.8	-25.5	0.4	-14.3	-22.6	0.5	-12.9	10.7	-2.6	9.7	-3.8	5.7	290	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	23.2	14.9	21.0	14.3	19.2	13.6	16.0	21.5	15.2	19.6	14.5	18.0	3.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	9.9	17.0	13.2	9.5	16.3	12.7	9.1	15.7	44.6	21.6	42.3	19.6	40.3	17.9	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.4	7.8	DB	-19.6	27.5	2.3	2.5	-21.2	29.3	-22.6	30.8	-23.9	32.2	-25.6	34.0	(4)
(5)			WB	-20.1	17.2	2.3	1.1	-21.7	18.0	-23.0	18.6	-24.3	19.2	-26.0	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-3.4	-2.5	0.0	5.1	10.1	13.3	14.5	13.9	10.5	5.8	-0.5	-3.0	(6)	
	DBStd	7.75	5.97	5.07	4.57	2.82	2.88	2.42	2.01	2.07	2.10	2.74	4.95	5.48	(7)	
	HDD10.0	2130	416	350	311	151	34	1	0	0	17	132	316	403	(8)	
	HDD18.3	4743	674	583	569	398	255	153	122	138	234	389	566	662	(9)	
	CDD10.0	434	0	0	0	3	37	100	139	122	33	1	0	0	(10)	
	CDD18.3	7	0	0	0	0	0	3	2	2	0	0	0	0	(11)	
	CDH23.3	66	0	0	0	0	2	27	18	18	0	0	0	0	(12)	
	CDH26.7	9	0	0	0	0	0	3	3	3	0	0	0	0	(13)	
(14) Wind	WSAvg	4.2	5.0	4.9	4.7	4.1	3.7	3.3	3.2	3.2	3.7	4.2	4.9	5.1	(14)	
(15) Precipitation	PrecAvg	1216	124	119	88	65	49	38	41	67	142	240	148	141	(15)	
	PrecMax	1517	276	263	167	156	183	133	85	179	309	517	348	325	(16)	
	PrecMin	915	26	0	0	9	5	5	3	0	9	103	37	18	(17)	
	PrecStd	203	63	62	49	43	41	28	20	44	84	106	81	73	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	7.0	11.0	19.4	23.0	26.7	25.8	25.8	19.9	12.8	9.4	8.5	(19)	
		MCWB	6.2	4.6	6.6	10.5	12.5	16.0	16.2	16.0	12.7	9.8	7.7	7.0	(20)	
	2%	DB	6.1	5.8	7.9	14.3	20.4	22.9	22.2	22.1	16.9	11.2	7.8	6.6	(21)	
		MCWB	5.0	4.2	4.7	7.5	11.7	14.6	15.7	15.0	11.7	9.0	6.3	5.5	(22)	
	5%	DB	4.8	4.7	6.3	11.7	18.1	20.0	20.2	19.6	14.9	10.3	6.7	5.3	(23)	
		MCWB	3.9	3.3	3.8	6.2	11.0	13.7	15.1	14.5	11.4	8.5	5.4	4.3	(24)	
	10%	DB	3.9	3.6	5.0	9.8	15.9	17.9	18.4	17.5	13.5	9.5	5.5	3.9	(25)	
		MCWB	3.1	2.4	3.0	5.6	10.2	13.0	14.4	13.9	11.2	8.0	4.3	3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	5.4	7.5	10.8	13.8	16.5	17.3	16.9	14.1	10.8	7.9	6.8	(27)	
		MCDB	7.4	6.4	10.0	18.3	20.6	24.8	22.7	24.1	16.8	11.8	9.2	7.9	(28)	
	2%	WB	5.1	4.5	5.5	8.1	12.6	15.3	16.3	15.7	13.0	9.9	6.7	5.7	(29)	
		MCDB	5.9	5.5	7.6	12.8	18.7	21.5	21.3	20.6	15.5	10.8	7.8	6.7	(30)	
	5%	WB	4.0	3.6	4.2	6.9	11.6	14.2	15.4	15.0	12.1	9.0	5.6	4.3	(31)	
		MCDB	4.8	4.5	5.7	10.5	16.8	18.9	19.5	18.6	14.1	10.0	6.6	5.1	(32)	
	10%	WB	3.1	2.6	3.4	6.0	10.6	13.3	14.6	14.2	11.5	8.0	4.4	2.9	(33)	
		MCDB	3.9	3.5	4.7	9.1	15.0	17.2	18.0	16.9	13.1	9.2	5.3	3.6	(34)	
(35) Mean Daily Temperature Range	MDBR		4.6	5.5	5.6	7.1	7.9	6.9	5.8	5.9	5.2	4.6	4.2	4.5	(35)	
	5% DB	MCDBR	4.4	5.5	6.8	11.3	13.0	12.5	10.8	11.4	8.8	5.4	4.6	4.8	(36)	
		MCWBR	3.8	4.2	4.4	6.3	6.5	5.9	5.3	5.5	4.9	3.7	3.7	4.1	(37)	
	5% WB	MCDBR	4.4	4.6	5.7	9.4	10.8	10.4	9.5	9.0	6.3	4.3	4.2	4.7	(38)	
		MCWBR	3.9	3.8	4.1	5.6	5.8	5.1	4.9	4.7	3.9	3.4	3.6	4.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.299	0.331	0.365	0.381	0.416	0.403	0.381	0.343	0.315	0.277	0.255	(40)	
	taud		1.997	2.104	2.036	1.999	2.070	2.087	2.208	2.291	2.360	2.217	2.022	1.960	(41)	
	Ebn at Noon		462	693	782	823	841	814	818	809	784	670	469	360	(42)	
	Edh at Noon		53	80	117	145	146	146	127	108	86	71	50	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.39	1.14	2.44	4.07	5.38	5.16	4.62	3.57	2.29	1.10	0.50	0.25	(44)	
	RadStd		0.04	0.17	0.27	0.41	0.48	0.34	0.39	0.38	0.26	0.14	0.06	0.02	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+0.82	+0.58	N/A	N/A	N/A	+5	(46)
(47)	Regional (10 neighbors)	N/A	N/A	+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page